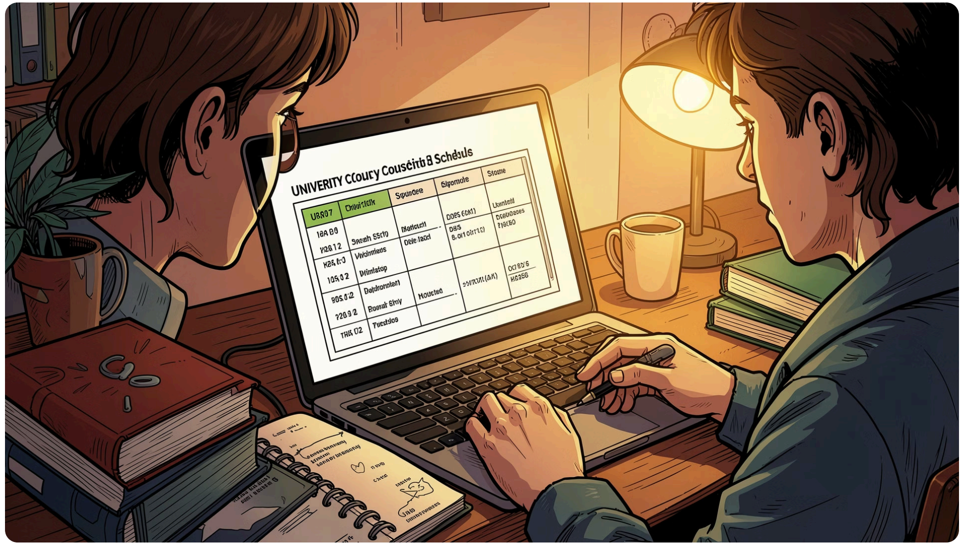
SOSE · Prof. Marco Autili · A.Y. 2025/2026 · Marco Spada · id: 308887

Ethics as a Service in SOA University Course Assignment

- **Application domain:** University services. A Service-Oriented Architecture (SOA) designed to support the assignment of courses to students.
- **Focus:** Moving beyond naive automation. Implementing an *Ethics by Design* approach to guarantee fair, transparent, and traceable decision-making.

Project Description & Core Philosophy



This project implements a Service-Oriented Architecture (SOA) that reframes university course assignment not merely as a technical automation, but as an **operational governance** challenge.

Acting as an **orchestrator**, the client coordinates the interaction between services, proving that data-driven decisions are not automatically neutral and require an explicit ethical validation layer.

Data as a Service (DaaS)

Decouples data management by exposing student and course semantic profiles via REST APIs, directly querying an underlying RDF Triplestore.

Ethics as a Service (EaaS)

Functions as a bounded service block that autonomously evaluates the assignment context against dynamic, externalized JSON policies.

System Goals & Architectural Objectives

Semantic Data Abstraction

Exposing university data as a reusable DaaS via REST APIs, hiding the complexity of underlying SPARQL queries from the client.

Agnostic Governance

Enforcing transparent rules loaded from external JSON files, decoupling the ethical policies from the core application logic.

Explainability

Moving away from "blind oracles." Every outcome includes a clear rationale, a computed risk level, and required actions.

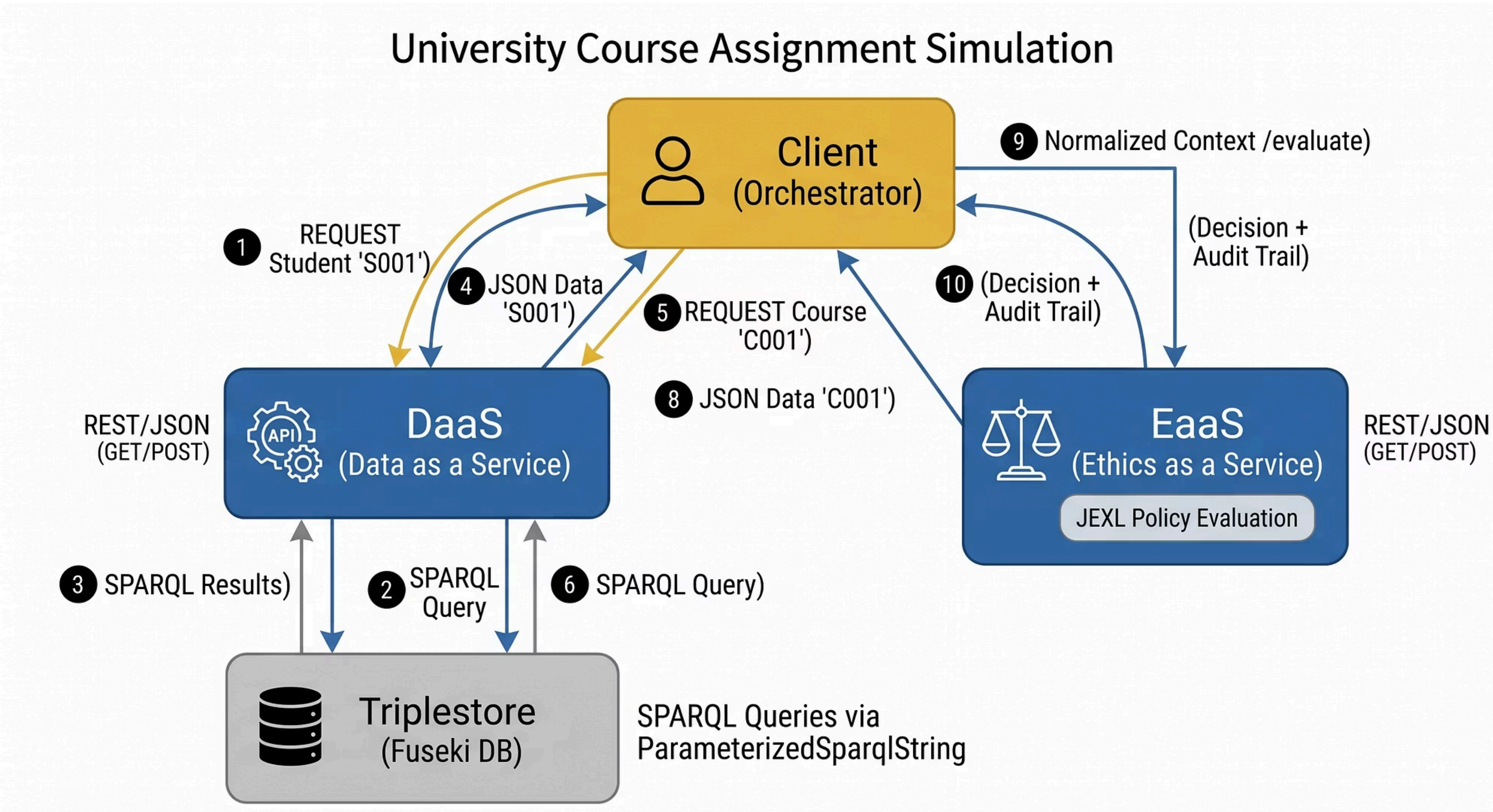
Immutable Traceability

Generating a persistent audit trail for every evaluation. Every pipeline stage leaves evidence for future accountability.

Human-in-the-Loop

Supporting human judgment rather than replacing it. The system automatically escalates high-risk or borderline cases to human tutors.

Service Architecture & Orchestration Topology



Separation of Concerns (SoC)

Complete decoupling between three autonomous tiers: Semantic Data (DaaS + RDF), Application Logic (Client), and Operational Governance (EaaS).

Orchestration Pattern

The Client acts as a centralized orchestrator, handling data retrieval, payload normalization, and ethical validation sequentially.

Component Interchangeability

Every microservice communicates via standard HTTP/JSON contracts, meaning any component can be replaced or updated without affecting the others.

Data as a Service: Semantic Layer & REST Federation

RDF Dataset: Schema & Vocabulary

Entity: foaf:Person (Student)

- ex:hasGPA (Academic performance tracking)
- ex:isWorkingStudent (Socio-economic context proxy)
- ex:needsAccessibility (Physical/logistic constraint)

Entity: ex:Course (Course)

- ex:hasPrerequisite (Graph-based dependency / propedeuticità)
- ex:workload (High/Low study-load classification)
- ex:requiresPhysicalPresence (Delivery mode constraint)

Exposed REST Endpoints (The Information Layer)

01

GET /api/courses

Full graph serialization to JSON.

02

GET /api/courses/{id}

Specific resource retrieval.

03

GET /api/courses/search

Multi-conditional parameterized SPARQL query.

04

GET /api/students/{id}

Student profile retrieval.

05

GET /api/docs

Interactive Swagger UI (OpenAPI v3 Contract).

Data Virtualization Merit: The client is completely decoupled from the Triplestore. It consumes clean, interoperable JSON payloads via standard HTTP, entirely shielded from underlying SPARQL query complexities and graph traversals.

Ethics as a Service: The Governance Pipeline

Core Logic & Architecture

Stateless Policy Engine

EaaS ingests a structured JSON context (Student + Course) and evaluates it against externalized policy rules via Apache JEXL.

Zero-Trust Principle

The service strictly rejects any pre-declared risk labels from the caller, eliminating the risk of client-side underestimation or manipulation.

EaaS is not a passive data filter, but a **Bounded Service Block** enforcing deterministic operational governance.

Structured Evaluation Output

- Computed Risk Level & Contextual Rationale
- Triggered Policy Bindings
- Required Mitigation Actions
- Immutable Audit Token & Trace Reference

PROCEED

Safe Flow: No policy thresholds crossed. Request is ethically compliant.

REVISE

Conditional Approval: Minor conflicts detected. Requires parameter optimization (e.g., workload reduction).

ESCALATE

Human-in-the-Loop: High-risk or borderline anomaly. Triggers an immediate handoff to a human tutor.

REJECT

Hard Policy Block: Critical ethical violation detected (e.g., accessibility breach). Operation halted.

Execution Scenario 1: Hard Policy Violation

Context Aggregation (Client-Side)

Student Profile (Alice)

- ex:needsAccessibility = true
- ex:isWorkingStudent = true
- Academic Merit: GPA 22.5/30

Course Resource (Algorithms)

- ex:requiresPhysicalPresence = true
- ex:workload = "High"

Triggered Policy Binding

POL-001 (Accessibility Conflict): A course requiring mandatory physical presence cannot be assigned to a student with documented accessibility constraints.

EaaS Operational Output

Decision

REJECT - Hard policy block enforced. The automated enrollment is immediately halted.

Rationale (Explanation)

Critical conflict detected: The course requires on-campus attendance, which directly violates the student's remote/accessible learning requirements.

Audit Trail (Traceability)

Unique UUID generated. The system stores the permanent proof of rejection, linking the outcome to POL-001 for future administrative compliance or human review.

Execution Scenario 2: Frictionless Compliance

Context Aggregation (Client-Side)

Student Profile (Bob)

- ex:needsAccessibility = false
- ex:isWorkingStudent = false
- Academic Merit: GPA 28.5/30

Course Resource (Human-Computer Interaction)

- ex:requiresPhysicalPresence = false
- ex:workload = "Low"
- ex:hasPrerequisite = false

Triggered Policy Binding

POL-004 (Standard Assignment): The default fallback policy is safely activated when no risk thresholds are exceeded.

EaaS Operational Output

Decision

PROCEED - Frictionless assignment allowed without requiring external or manual interventions.

Risk Level

LOW - No ethical alert conditions detected within the aggregated input context.

Continuous Auditing (Transparency)

An immutable UUID is generated regardless of the positive outcome. Every decision leaves a permanent trace to prove that ethical safety was actively verified, not merely assumed.

Execution Scenario 3: Ethical Dilemma & Human Handoff

Context Aggregation (Client-Side)

Student Profile (Charlie)

- ex:isWorkingStudent = true
- ex:needsAccessibility = false
- Academic Risk: GPA 19.0/30 (Below safety threshold).

Course Resource (Artificial Intelligence)

- ex:workload = "High"
- ex:hasPrerequisite = true

Triggered Policy Binding

POL-003 (Student Protection): Triggered when a student in academic difficulty (Low GPA) attempts to enroll in a high-workload/high-complexity course.

EaaS Operational Output

Decision

ESCALATE - The system refuses to take an autonomous "Yes/No" decision. The process is paused for human expert review.

Rationale (Explanation)

The combination of a high-complexity course and academic vulnerability requires a qualitative interview with a human tutor to assess student motivation and feasibility.

Audit & Responsibility

Required Action: **ASSIGN_ACADEMIC_TUTOR**. The audit trail records this escalation, ensuring that the "responsibility gap" is filled by a designated human operator rather than a rigid algorithm.

Short Critical Reflection: The Limits of EaaS

Context Blindness

Algorithms cannot grasp human nuances. A system can only evaluate what is mapped within structured data (RDF). Unquantifiable hardships or temporary personal contexts remain invisible to the automated logic.

Policy Bias

Governance rules are authored by humans. If these external policies are not regularly reviewed, they risk crystallizing systemic prejudices or enforcing rigid paternalism under the guise of "ethics."

Illusion of Provenance

An ethical decision is only as reliable as the accuracy of its input data. If the DaaS provides outdated or incorrectly classified semantic profiles, the resulting EaaS decision will be mathematically flawless but ethically unjust.

> Conclusion: No decision that definitively blocks a student's academic path or rights should be fully automated. The EaaS architecture is designed to be a decision-support pipeline, not a moral oracle. It must automate the safe cases (PROCEED), but fundamentally, it must recognize its own boundaries and proactively delegate borderline cases (ESCALATE) to human responsibility.